

Data Analytics Capstone

Air Quality Risk Prediction Using Environmental and Meteorological Data

Synopsis

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Introduction

Background and Context

Both completely natural as well as man made, poor air quality can have very detrimental effects on the conditions of the environment as well as the health of the public. This makes it an important issue in the context of Climate and Sustainability. Even recent studies noted that air pollution, in particular fine particulate matter such as PM_{2.5}, can be continuously linked to major health risks which means that it remains a central concern in the management and monitoring of the environment (Wan Mahiyuddin et al., 2023). The growing need for forward looking tools rather than backdated summaries of air pollution conditions by governments and communities has also caused the research to expand rapidly (Wang et al., 2025).

Many influences exist that can affect air quality, such as pollutants in the environment, temperature as a result of meteorological variables, the speed of the wind, levels of humidity and precipitation. Location and time varies greatly between these factors, and therefore anticipating poor air quality conditions in the absence of a structured predictive framework becomes extremely difficult and unreliable. Pollution accumulation can also be intensified or suppressed by seasonal effects. Therefore the outcome of interest is not random and predictive analytics becomes a very suitable approach for the domain of air quality as the outcome emerges from measurable patterns found in the environmental and weather conditions (Eren et al., 2025).

This topic becomes relevant in its support when predicting poor air quality leads to more informed environmental management, earlier warnings, and better monitoring which aids planners, sustainability teams, and environmental agencies in their decision making. It therefore has weighted practical value as these stakeholders gain benefit from data driven and timely

insight into the condition in which air quality is most likely to deteriorate. Therefore decision making switches from being reactive observation to actual proactive environmental management. The usefulness of such models for air quality management and early warning systems is emphasized in studies comparing machine learning approaches for AQI prediction (Tırink, 2025).

Problem Statement

Predicting poor air quality days consistently remains very difficult because of the influence of multiple interacting pollutant and meteorological factors. These factors are fluid and change over time, therefore without a predictive framework that is clear and precise identification of the main drivers of poor air quality, recognition of how seasonal variation plays a role, and ultimately selecting suitable models for the binary classification of poor air quality becomes extremely challenging for decision makers. There is a continuous note of absence of universally standardized evaluation practices in the literature, as well as a need for models that can effectively balance out predictive performance with its practical utility and interpretability (Zhou et al., 2024).

Purpose of the Study

The purpose of this study is to develop a predictive framework for the binary classification of poor air quality using environmental and meteorological data. More specifically, the study aims to identify the key variables associated with poor air quality days, examine whether those days can be accurately classified using pollutant and weather data, assess whether poor air quality days vary across seasons and time periods, and compare predictive model performance. Research trends towards increasingly combining exploratory analysis with

comparative model evaluation in air quality forecasting studies is consistent with the purpose of this study (Wang et al., 2025; Tırink, 2025).

Scope and Objectives

The scope of this study is limited to the analysis of poor air quality days using publicly available environmental and meteorological data. The study will specifically focus on Kapolei, Honolulu County, Hawaii, USA, and will cover the period from January 1, 2024 to December 31, 2025. The final analytical dataset will be structured at that daily level, with each row representing one calendar date for the selected study setting. The scope is intentionally restricted to keep the project feasible within the 10 week capstone timeline while still allowing for descriptive, inferential, and predictive analysis.

The study will use two public data sources: U.S. EPA AirData / Air Quality System (AQS) daily data and NOAA Climate Data Online (CDO) Daily Summaries for meteorological variables. These datasets will be merged into one final daily analytical file in order to examine how pollutant and weather conditions relate to air quality risk. The primary outcome of the study will be the binary target variable `poor\_air\_quality` derived from `overall\_aqi`, which will classify each day as either a poor air quality day or not a poor air quality day based on the selected study threshold. Therefore, the study is specifically designed around the binary classification of poor air quality, rather than broader multi-category AQI prediction.

Developing a predictive framework for the binary classification of poor air quality using environmental and meteorological data will serve as the main objective of this study. A related objective is to identify which pollutant and weather variables are most strongly associated with poor air quality days, so that the final framework is not only predictive but also interpretable. Another objective is to examine whether poor air quality days vary significantly across seasons

and other related time based groupings. The final objective is to compare candidate classification models in order to determine which model performs best for identifying poor air quality days within the selected study setting and period.

The following research questions will specifically guide this study:

- Research Question 1 (RQ1): Which environmental and meteorological variables are most strongly associated with poor air quality days?
- Research Question 2 (RQ2): Can poor air quality days be accurately identified and classified using pollutant and weather data?
- Research Question 3 (RQ3): Do poor air quality days vary significantly across seasons and time periods?
- Research Question 4 (RQ4): Which predictive model performs best for the binary classification of poor air quality?

Sample Size Calculation

Using either power based or confidence interval based methods, the sample size for each research question will be estimated separately. The largest of the necessitated minimum samples sizes will be selected as the final study sample size.

Table 1*Methods Used to Calculate Sample Sizes*

Research Question	Method Used	Key Parameters	Minimum Sample Size (N)
RQ1	Power/Green's Rule for multiple regression.	$\alpha = 0.05$, predictors = 8	114
RQ2	Confidence interval for proportion.	$Z = 1.96$, $e = 0.05$, $p = 0.50$	385
RQ3	Power for one way ANOVA.	$\alpha = 0.05$, power = 0.80, effect size $f = 0.25$, groups = 4	180
RQ4	Confidence interval for expected model accuracy.	$Z = 1.96$, $e = 0.05$, expected accuracy = 0.80	246

RQ1

Green's rule for multiple regression:

$$n \geq 50 + 8m$$

Where:

$$m = 8$$

So:

$$n \geq 50 + 8(8)$$

$$n \geq 50 + 64$$

$$n \geq 114$$

Thus:

$$n = 114$$

RQ2

Confidence interval formula for a proportion:

$$n = \frac{Z^2 p(1-p)}{e^2}$$

Substitute the values:

$$n = \frac{(1.96)^2 (0.50)(1-0.50)}{(0.05)^2}$$

$$n = \frac{(1.96)^2 (0.50)(0.50)}{(0.05)^2}$$

$$n = \frac{3.8416 \times 0.25}{0.0025}$$

$$n = \frac{0.9604}{0.0025}$$

$$n = 384.16$$

Rounded up:

$$n = 385$$

RQ3

One way ANOVA, using:

$$\alpha = 0.05$$

$$\text{Power} = 0.80$$

$$f = 0.25$$

$$k = 4$$

Estimated minimum sample size:

$$n \approx 180$$

Compact sample size:

$$n = 180$$

Please note that the RQ3 estimate is presented as an approximate power based planning value for the one-way ANOVA under the stated assumptions.

RQ4

Confidence interval formula for expected model accuracy:

$$n = \frac{Z^2 p(1-p)}{e^2}$$

Substitute the values:

$$n = \frac{(1.96)^2 (0.80)(1-0.80)}{(0.05)^2}$$

$$n = \frac{(1.96)^2 (0.80)(0.20)}{(0.05)^2}$$

$$n = \frac{3.8416 \times 0.16}{0.0025}$$

$$n = \frac{0.614656}{0.0025}$$

$$n = 245.86$$

Rounded up:

$$n = 246$$

Please note that the RQ4 estimate is presented as a planning orientated confidence interval approach to expected model accuracy and serves as a practical minimum sample benchmark for model evaluation.

Final maximum sample size

$$\max(114, 385, 180, 246) = 385$$

Minimum required sample size:

$$n = 385$$

In reference to the calculations and Table 1, 385 observations is the minimum sample size necessary. This study will therefore aim to use at least 730 daily observations in order to allow for any missing values, seasonal analysis, and model training and testing.

Data Description

This study will use two publicly available datasets: U.S. EPA AirData / Air Quality System (AQS) daily data and NOAA Climate Data Online (CDO) Daily Summaries. Together these sources provide the environmental and meteorological variables needed to support the binary classification of poor air quality. Kapolei, Honolulu County, Hawaii, USA will serve as the setting for this study and January 1, 2024 to December 31, 2025 will serve as the study period. The final analytical dataset will be structured at the daily level, with each row representing one calendar date for the selected study setting.

EPA AirData / Air Quality System (AQS) daily data will serve as the primary air quality source. This dataset provides daily pollutant and AQI related measures collected from outdoor monitoring systems in the United States. The EPA data will provide pollutant variables such as PM2.5, PM10, ozone, and nitrogen dioxide, as well as pollutant specific AQI values and selected daily observation quality indicators. These variables will be utilized for this study and are appropriate because they support both explanatory analysis and the construction of the study outcome. The EPA data source is available at: <https://www.epa.gov/outdoor-air-quality-data>.

The NOAA Climate Data Online Daily Summaries dataset will provide the meteorological variables used in the study. These variables include daily maximum temperature, daily minimum temperature, precipitation, and average wind speed where available. Meteorological conditions are relevant to this study because they can influence pollutant

accumulation, dispersion, and daily changes in air quality. The NOAA data source is available at: <https://www.ncei.noaa.gov/cdo-web/>.

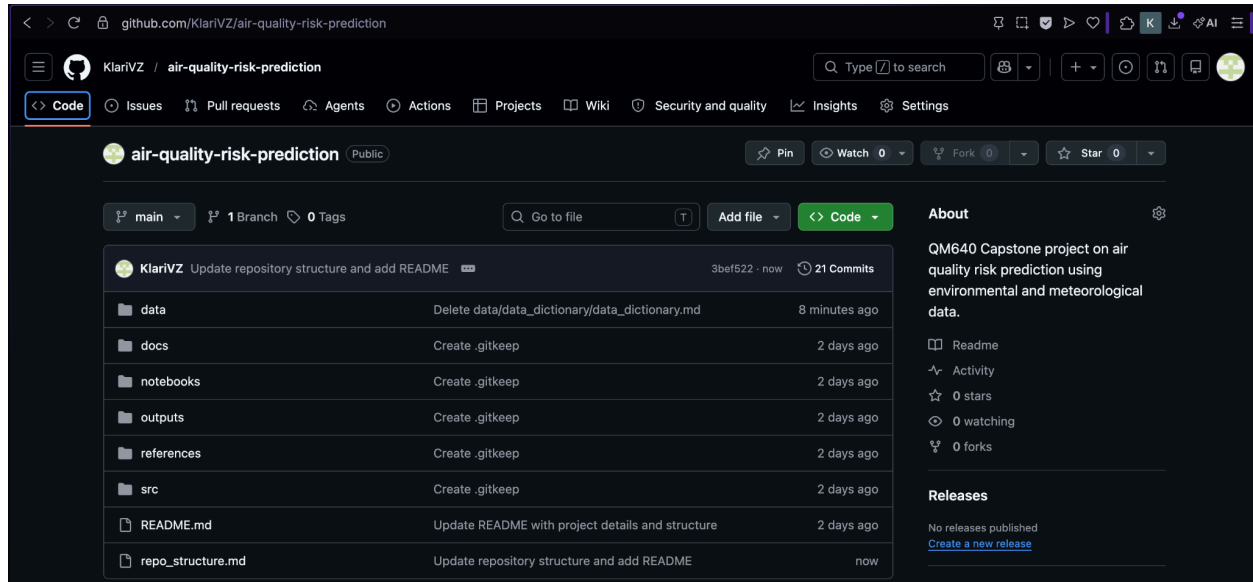
The final analytical dataset will be created by merging the EPA and NOAA data by date and by the selected study location or nearest appropriate reporting station. After merging, the dataset will contain both environmental and meteorological predictors for the same daily observation period. Prior to analysis, the dataset will be cleaned by standardizing variable formats, and converting NOAA raw weather values into cleaned reporting units where required. This process is necessary to ensure that the final daily dataset is analytically consistent and suitable for descriptive statistics, inferential testing, and predictive modeling.

The primary study outcome will be `'poor_air_quality'`, a binary target variable derived from `'overall_aqi'`. The variable `'overall_aqi'` will be calculated as the maximum AQI value across the pollutant specific AQI measures available for a given date. Based on the selected study threshold, the binary target will then classify each day as either a poor air quality day or not a poor air quality day.

A public GitHub repository titled `'air-quality-risk-prediction'`, has already been created for this project and is available at <https://github.com/KlariVZ/air-quality-risk-prediction>. The current repository structure includes folders for raw and processed data, documentation, notebooks, outputs, references, source code, and the project README. This structure supports the storage of raw source files, cleaned analytical data, figures, tables, scripts, and related project materials in one accessible location.

Figure 1

Screenshot of GitHub repository overview for air quality risk prediction.



Note: The repository structure currently includes the following major folders: `data/`, `docs/`, `notebooks/`, `outputs/`, `references/`, and `src/`, with subfolders for the data dictionary, raw data, processed data, figures, models, and tables. This structure will help keep the project files organized and accessible for evaluation.

Figure 2

Screenshot of `repo\_structure.md` file contents showing GitHub repository folder and file structure for air quality risk prediction.

```

1  # Current GitHub Folder Structure for `air-quality-risk-prediction` Repository
2
3  air-quality-risk-prediction/
4  |
5  |   data/
6  |   |
7  |   |   data_dictionary/
8  |   |   |
9  |   |   |   data_dictionary.csv
10 |   |   |
11 |   |   |   processed/
12 |   |   |   |
13 |   |   |   |   .gitkeep
14 |   |   |   |
15 |   |   |   |   raw/
16 |   |   |   |   |
17 |   |   |   |   |   raw_air_quality_pm25.csv
18 |   |   |   |   |   raw_air_quality_pm10.csv
19 |   |   |   |   |   raw_air_quality_ozone.csv
20 |   |   |   |   |   raw_air_quality_no2.csv
21 |   |   |   |   |   raw_weather_noaa.csv
22 |   |   |   |
23 |   |   |   |   docs/
24 |   |   |   |   |
25 |   |   |   |   |   .gitkeep
26 |   |   |   |   |
27 |   |   |   |   |   notebooks/
28 |   |   |   |   |   |
29 |   |   |   |   |   |   .gitkeep
30 |   |   |   |   |   |
31 |   |   |   |   |   |   outputs/
32 |   |   |   |   |   |   |
33 |   |   |   |   |   |   |   figures/
34 |   |   |   |   |   |   |   |
35 |   |   |   |   |   |   |   |   .gitkeep
36 |   |   |   |   |   |   |   |
37 |   |   |   |   |   |   |   |   models/
38 |   |   |   |   |   |   |   |   |
39 |   |   |   |   |   |   |   |   |   .gitkeep
40 |   |   |   |   |   |   |   |   |
41 |   |   |   |   |   |   |   |   |   tables/
42 |   |   |   |   |   |   |   |   |   |
43 |   |   |   |   |   |   |   |   |   |   .gitkeep
44 |   |   |   |   |   |   |   |   |   |
45 |   |   |   |   |   |   |   |   |   |   references/
46 |   |   |   |   |   |   |   |   |   |   |
47 |   |   |   |   |   |   |   |   |   |   |   .gitkeep
48 |   |   |   |   |   |   |   |   |   |   |
49 |   |   |   |   |   |   |   |   |   |   |   src/
50 |   |   |   |   |   |   |   |   |   |   |   |
51 |   |   |   |   |   |   |   |   |   |   |   |   .gitkeep
52 |   |   |   |   |   |   |   |   |   |   |   |
53 |   |   |   |   |   |   |   |   |   |   |   |   README.md
54 |   |   |   |   |   |   |   |   |   |   |   |   repo_structure.md
55 |   |   |   |   |   |   |   |   |   |   |   |
56 |   |   |   |   |   |   |   |   |   |   |   |   .gitkeep
57 |   |   |   |   |   |   |   |   |   |   |   |
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99 |   |   |   |   |   |   |   |   |   |   |   |   .gitkeep
100 |   |   |   |   |   |   |   |   |   |   |   |   .gitkeep

```

Note: The `repo\_structure.md` file will be updated as the project progresses.

Data Dictionary

Table 2 presents the planned data dictionary for the final cleaned and merged analytical dataset.

Table 2*Planned Data Dictionary for the Final Analytical Dataset*

Variable Name	Description	Source	Data Type	Unit of Measurement	Role in Study
date	Calendar date of observation	EPA / NOAA	Date	YYYY-MM-DD	Merge key / Identifier
site_id	Unique EPA monitoring site identifier for the Kapolei monitoring location	EPA	String / Integer	Not applicable	Identifier
local_site_name	Name of the air monitoring site	EPA	String	Not applicable	Identifier
county	County in which the monitoring site is located	EPA	String	Not applicable	Identifier
state	State in which the monitoring site is located	EPA	String	Not applicable	Identifier
site_latitude	Latitude of the EPA monitoring site	EPA	Float	Decimal degrees	Site metadata
site_longitude	Longitude of the EPA monitoring site	EPA	Float	Decimal degrees	Site metadata

daily_max_1 hr_no2	Daily maximum 1-hour nitrogen dioxide concentration	EPA	Float	ppb	Predictor
daily_max_8 hr_ozone	Daily maximum 8-hour ozone concentration	EPA	Float	ppm	Predictor
daily_mean_ pm25	Daily mean PM2.5 concentration	EPA	Float	ug/m3	Predictor
daily_mean_ pm10	Daily mean PM10 concentration	EPA	Float	ug/m3	Predictor
daily_aqi_no 2	Daily AQI value associated with NO2	EPA	Integer	AQI index	Intermediate variable
daily_aqi_oz one	Daily AQI value associated with ozone	EPA	Integer	AQI index	Intermediate variable
daily_aqi_pm 25	Daily AQI value associated with PM2.5	EPA	Integer	AQI index	Intermediate variable
daily_aqi_pm 10	Daily AQI value associated with PM10	EPA	Integer	AQI index	Intermediate variable
daily_obs_co unt_no2	Number of observations used to derive daily NO2	EPA	Integer	Count	Data quality variable

	summary value				
daily_obs_count_ozone	Number of observations used to derive daily ozone summary	EPA	Integer	Count	Data quality variable
daily_obs_count_pm25	Number of observations used to derive daily PM2.5 summary value	EPA	Integer	Count	Data quality variable
daily_obs_count_pm10	Number of observations used to derive daily PM10 summary value	EPA	Integer	Count	Data quality variable
percent_complete_no2	Percentage completeness of NO2 observations for the day	EPA	Float	Percent	Data quality variable
percent_complete_ozone	Percentage completeness of ozone observations for the day	EPA	Float	Percent	Data quality variable
percent_complete_pm25	Percentage completeness of PM2.5 observations for the day	EPA	Float	Percent	Data quality variable
percent_complete	Percentage completeness of	EPA	Float	Percent	Data quality

plete_pm10	PM10 observations for the day				variable
station	NOAA station identifier for the selected weather station	NOAA	String	Not applicable	Identifier
station_name	Name of the selected NOAA weather station	NOAA	String	Not applicable	Site metadata
latitude	Latitude of the NOAA weather station	NOAA	Float	Decimal degrees	Site metadata
longitude	Longitude of the NOAA weather station	NOAA	Float	Decimal degrees	Site metadata
elevation	Elevation of the NOAA weather station	NOAA	Float	Meters	Site metadata
tmax	Daily maximum temperature	NOAA	Float	Degrees Celsius	Predictor
tmin	Daily minimum temperature	NOAA	Float	Degrees Celsius	Predictor
prcp	Daily precipitation	NOAA	Float	Millimetres	Predictor
awnd	Average daily wind speed	NOAA	Float	Meters per second	Predictor

year	Year extracted from date	Derived	Integer	Year	Time feature
month	Month extracted from date	Derived	Integer	1 to 12	Time feature
season	Season category assigned to the observation date	Derived	String	Not applicable	Time feature
day_of_week	Day of the week derived from date	Derived	String / Integer	Not applicable	Time feature
is_weekend	Indicator showing whether the observation date falls on a weekend	Derived	Binary	0 = No, 1 = Yes	Time feature
overall_aqi	Daily overall AQI calculated as the maximum AQI across pollutant specific AQI values	Derived	Integer	AQI index	Outcome construction
poor_air_quality	Binary target variable indicating whether the day is classified as a poor air quality day	Derived	Binary	0 = Not poor, 1 = Poor	Target variable

Analytic Approach

The analytical approach for this study will combine descriptive statistics, inferential statistics, and predictive modelling in order to answer the four chosen research questions in a structured and feasible way. Because the study is not limited to prediction alone the combination

of this approach is appropriate. It also seeks to identify the variables most strongly associated with poor air quality days, examine whether poor air quality days vary across seasons and time periods, and compare the performance of multiple classification models. Therefore the use of both traditional statistical methods and machine learning approaches within a single analytical framework becomes necessary.

Data preparation and exploratory data analysis (EDA) will be the starting point of the analysis. This stage will include checking for missing values, verifying date alignment between the EPA and NOAA datasets, reviewing duplicate dates, assessing data completeness, and converting meteorological variables into standard reporting units where necessary. Descriptive statistics such as the mean, median, standard deviation, minimum, maximum, and interquartile range will be calculated for the main pollutant and weather variables. Visual exploration will include line, histograms, boxplots, and correlation heatmaps to understand distributions, identify outliers, and examine temporal patterns. The quality and structure of the final daily dataset will guide the selection of the most appropriate inferential tests and predictive models, which makes this stage extremely important.

Because the overall target variable for the study is `'poor_air_quality'`, derived from `'overall_aqi'`, the central modeling task is a binary classification problem. However, some of the exploratory and temporal research questions also necessitate inferential statistical testing. The study will therefore use statistical methods for hypothesis testing and machine learning methods for classification and model comparison. The analysis will be conducted at the daily observation level, with each row in the final analytical dataset representing one calendar date for the selected Kapolei study setting.

Hypothesis tests will be evaluated at the 5% significance level ($\alpha = 0.05$) for the inferential analysis. Where assumptions of normality or homogeneity of variance are not met sufficiently, appropriate nonparametric alternatives will be considered. For predictive modeling, the dataset will be partitioned into training and test data, and cross-validation will be used where feasible to reduce overfitting and improve the reliability of the model comparison. Since this project is intended to remain feasible within a 10 week capstone timeline, the study will prioritize models that are widely accepted, interpretable, and manageable for a daily dataset of this size.

The selected evaluation metrics for the classification component are accuracy, precision, recall, F1-score, and ROC-AUC. No single metric is sufficient for evaluating a poor air quality classifier, therefore the mentioned metrics were selected. Accuracy provides an overall measure of correct classification, but it could be misleading if the classes are imbalanced. Precision indicates how many days predicted as poor air quality were actually poor air quality days, which makes it useful. Recall has special importance as it reflects how effectively the model identifies true poor air quality days, which is highly relevant in an early warning or monitoring context. F1-score balances precision and recall and is therefore useful when both false positives and false negatives matter, which is why it is included. ROC-AUC measures the model's ability to distinguish between classes across classification thresholds rather than at only one cutoff point, which makes it an appropriate choice. In combination, these metrics provide a more complete and defensible assessment of model performance.

For this study, logistic regression, decision tree, random forest, and support vector machine have been chosen as the candidate machine learning models. The choice of these models hinged on the fact they provide a balance between interpretability, predictive power, and

feasibility. As a baseline statistical classifier, logistic regression is appropriate because it is interpretable and allows direct examination of the relationship between predictor variables and the probability of poor air quality. Decision trees are useful because they are easy to interpret and can model nonlinear decision rules. Random forest can improve predictive performance by combining multiple trees and can also provide feature importance measures, which is why it is included. Support vector machine is included because it is effective for classification problems involving potentially complex boundaries between classes. The fourth research question justifies the use of multiple models, which requires a comparison of predictive performance rather than reliance on a single method.

This analytic design allows the study to address explanation, classification, seasonal variation, and model comparison within a single coherent framework.

Solution to RQ1-4

RQ1: Which environmental and meteorological variables are most strongly associated with poor air quality days?

The study will make use of a combination of descriptive statistics, correlation analysis, and binary logistic regression to address RQ1. Firstly, descriptive statistics will be used to summarize the main pollutant and meteorological variables. Thereafter, correlation analysis will be utilized as an exploratory step to assess the strength and direction of pairwise relationships among the continuous predictor variables. As the outcome of the main study is the binary variable 'poor_air_quality', the principal inferential method for RQ1 will be binary logistic regression, which is appropriate when the dependent variable has two categories. Logistic regression allows the study to estimate the association between each environmental or

meteorological predictor and the probability of a poor air quality day while holding the other predictors constant, which makes it very suitable in this case.

Before fitting the final logistic regression model, the study will check for multicollinearity among predictors, using methods such as correlation review and variance inflation factor (VIF), in order to reduce instability in coefficient estimates. The Wald test for regression will be used to assess the significance of individual predictors, while the overall usefulness of the model will be reviewed using model fit measures appropriate to logistic regression. In addition to the regression results, random forest feature importance may be used as a supplementary interpretive tool to compare variable influence from a machine learning perspective. This combination aids both statistical inference and interpretation at a practical level.

H_{01} : Environmental and meteorological variables are not significantly associated with poor air quality days.

H_{a1} : At least one environmental or meteorological variable is significantly associated with poor air quality days.

RQ2: Can poor air quality days be accurately identified and classified using pollutant and weather data?

The study will frame poor air quality identification as a supervised binary classification problem to address RQ2. The candidate models will be logistic regression, decision tree, random forest, and support vector machine. The practical range of approaches from interpretable statistical classification to more flexible machine learning classification informed the selection of these models. Logistic regression will serve as a strong baseline because it is transparent and easy to interpret. Decision tree will allow simple rule based classification and is useful for

explaining prediction logic. Random forest will be used as it can capture nonlinear interactions and tends to perform well in environmental classification tasks. Support vector machine can perform strongly when the separation between classes is not purely linear, which is why it will be tested.

The dataset will be divided into training and test sets, and cross-validation will be applied during model development where feasible. The performance of the model will be assessed using accuracy, precision, recall, F1-score, and ROC-AUC on the test data. The goal is not merely to maximize overall correctness, but also to evaluate how effectively the model identifies true poor air quality days, which makes these metrics appropriate. Recall is particularly important because failing to identify a genuinely poor air quality day could be more problematic than incorrectly flagging a day as poor. The balance of tradeoff between precision and recall is where F1-score will be useful, while ROC-AUC evaluates discrimination across thresholds.

Model results will be compared against baseline performance to determine whether classification performance is meaningfully better than chance. The final interpretation for this research question will focus on whether pollutant and weather data provide sufficient predictive value to support reliable binary classification of poor air quality days.

H_{02} : Poor air quality days cannot be classified using pollutant and weather data better than chance.

H_{a2} : Poor air quality days can be classified using pollutant and weather data significantly better than chance.

RQ3: Do poor air quality days vary significantly across seasons and time periods?

The study will examine whether poor air quality days differ across seasonal and temporal groupings, such as season, month, or weekend versus weekday, in order to address RQ3. Since

differences across groups are the focus of this research question, the analysis will combine descriptive visualization with inferential group comparison tests.

When comparing the mean overall AQI across more than two groups, such as the four seasons, the primary statistical test will be one-way ANOVA, provided the assumptions of approximate normality and homogeneity of variance are reasonably satisfied. If ANOVA shows statistically significant results, post hoc comparisons, such as Tukey's Honest Significant Difference test, will be utilized to identify which groups differ from one another. In the case of comparing only two groups, such as weekday versus weekend, the study will use an independent samples t-test where assumptions are met. If the assumptions for parametric testing are not met, nonparametric alternatives such as the Kruskal-Wallis test or Mann-Whitney U test may be used instead.

Seeing as the study also uses a binary target variable, seasonal variation in the proportion of poor air quality days may additionally be assessed using a chi-square test of independence, where appropriate, to examine whether the frequency of poor air quality days differs significantly across categorical time groupings. This strengthens the analysis by aligning the statistical test with the form of the variable being evaluated.

H_{03} : There are no statistically significant differences in poor air quality days across seasons and time periods.

H_{a3} : Poor air quality days differ significantly across seasons and time periods.

RQ4: Which predictive model performs best for the binary classification of poor air quality?

Predictive performance of the selected classification models will be compared using a common evaluation framework in order to address RQ4. Once again, candidate models will

include logistic regression, decision tree, random forest, and support vector machine. In order to ensure fair comparison, each model will be trained using the same predictor set and evaluated under the same train/test and cross validation strategy.

Accuracy, precision, recall, F1-score, and ROC-AUC will be the main comparison criteria. The best performing model will not be identified on the basis of accuracy alone. Instead, overall balance of predictive performance, the ability to detect poor air quality days, and the practical interpretability of the model will be considered for the final selection of the model. This is important because in an environmental monitoring context, a slightly more interpretable model may be preferable if its performance is competitive and easier to explain to relevant stakeholders.

Model comparison may also include a formal statistical comparison of classifier predictions on the same test set using McNemar's test where it is deemed appropriate, especially when comparing two classifiers directly in terms of paired classification outcomes. This provides a more rigorous basis for evaluating whether observed differences in classification performance are likely to reflect a real performance difference rather than random variation in the test sample. The final conclusion for RQ4 will therefore consider both practical performance metrics and the statistical evidence supporting comparative performance.

H_{04} : There is no significant difference in predictive performance among the selected models for the binary classification of poor air quality.

H_{a_4} : At least one predictive model performs significantly better than the others for the binary classification of poor air quality.

Table 3

The primary evaluation metrics for the classification component.

Accuracy	$\text{Accuracy} = \frac{TP+TN}{TP+TN+FP+FN}$
Precision	$\text{Precision} = \frac{TP}{TP+FP}$
Recall	$\text{Recall} = \frac{TP}{TP+FN}$
F1-score	$F_1 = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$
ROC-AUC	$TPR = \frac{TP}{TP+FN}$ $FPR = \frac{FP}{FP+TN}$ ROC-AUC = Area under the ROC curve

Recommendation and Application

The expected outcome of this study is a practical and interpretable predictive framework for the binary classification of poor air quality using environmental and meteorological data. If acceptable predictive performance is demonstrated by the models, the study could provide evidence that pollutant and weather variables can be used to identify poor air quality days in a more timely and structured manner. This has practical value because it supports the movement from retrospective environmental observation toward more proactive environmental monitoring and risk awareness. Decision makers may use predictive insight to better anticipate when air quality is likely to deteriorate, rather than depending only on ‘in hindsight’ summaries of air quality conditions.

Primarily, the recommendation of this study is that integrating data driven classification tools should be considered as part of the decision support process toolkit for organizations involved in environmental monitoring and sustainability planning. In this study, the goal is not merely to determine whether poor_air_quality derived from overall_aqi can be predicted, but also to identify which environmental and meteorological variables are most strongly associated with those poor air quality days. This means the study can both support prediction and interpretability. From a practical perspective, this is important because decision makers often need not only an output, but also a clearer understanding of the conditions under which air quality is more likely to worsen.

Environmental protection agencies, local government planners, sustainability teams, and public information units serve as the main target users of the proposed solution. Environmental agencies may use the findings to strengthen local monitoring strategy, identify periods of higher risk, and improve situational awareness. Local government planners may use the results to better understand how environmental and weather conditions relate to local air quality risk patterns over time. Sustainability teams may use the findings to support internal environmental reporting, local planning discussions, or broader resilience related initiatives. Public information units may benefit from a simplified classification framework that can help communicate elevated air quality risk more clearly to relevant communities.

A secondary recommendation is that pollutant and meteorological variables should be viewed as jointly important rather than separately important by decision makers. One of the contributions of this study is that it brings both sources of information together into a single daily analytical framework. Poor air quality does not emerge from one factor alone, which makes this useful. Instead, it reflects the interaction of pollutant conditions and meteorological patterns

across time. The study may help organizations prioritize the factors they monitor most closely and better understand the environmental conditions associated with deterioration in air quality by identifying which variables most strongly contribute to the binary classification of poor air quality.

From an application perspective, the final model from this study could potentially be used as the analytical basis for a simple decision support tool or dashboard. A potential future implementation could accept daily pollutant and weather inputs and classify whether the day is likely to be a poor air quality day or not. The capstone may not include full deployment, but the study can still act as a demonstration of the practical pathway from raw environmental data to a more accessible and actionable risk classification output. There is value for potential users in a clear classification outcome, rather than multiple raw pollutant readings viewed in isolation.

Value in terms of resource prioritization and planning may also be offered by this study. If specific seasons, weather patterns, or pollutant combinations are found to be associated with poor air quality days, that information could be used by organizations to focus communication, monitoring attention, or preventive action more effectively. In this way, academic model comparison does not limit this study, rather, it also supports more informed environmental decision making by showing how predictive analytics can be translated into a practical monitoring context.

In summary, the recommendation of this study is that data driven binary classification of poor air quality should be considered a useful support tool for local environmental monitoring and sustainability decision making. The project is expected to provide practical value by identifying key drivers of poor air quality, testing whether poor air quality days can be classified with acceptable performance, examining temporal variation, and comparing models to determine

which approach is most suitable. Albeit only the scope of a capstone project, this creates a useful foundation for future environmental monitoring tools, local dashboards, or early warning support applications.

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